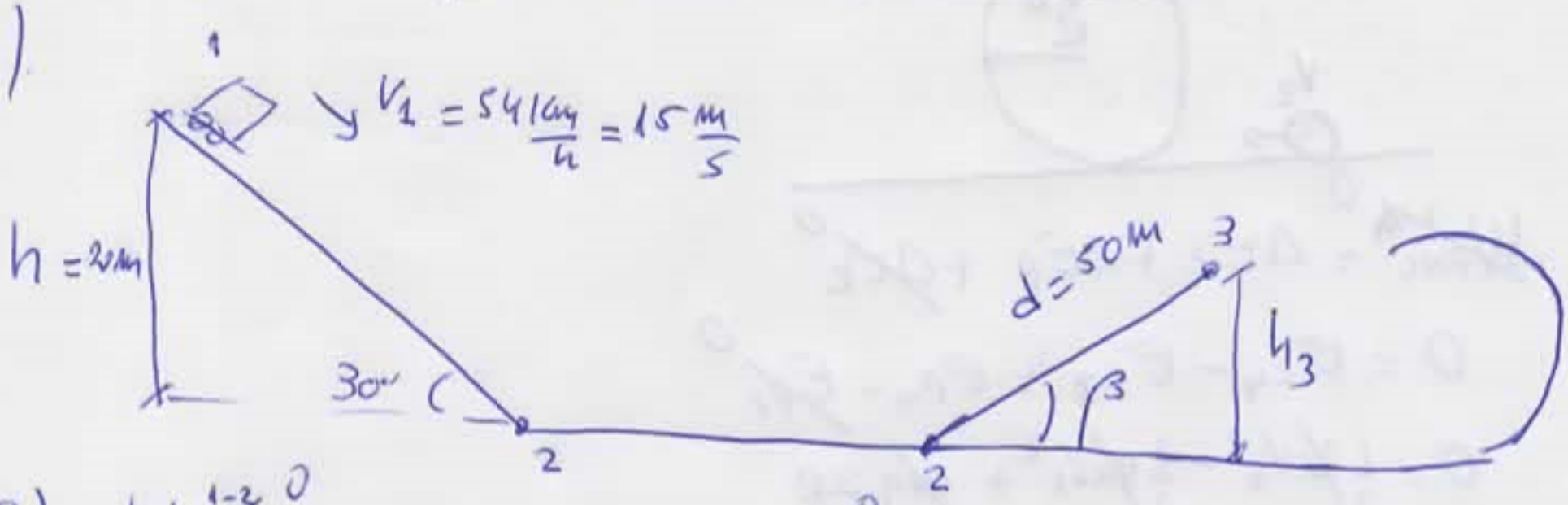
2° PARCIAL _ 1° FECHA _ M 11

1)



a) $W_{FNC}^{1-2} = \Delta E_c + \Delta E_p + \Delta E_k$ (0)

$0 = E_{c2} - E_{c1} + E_{p2} - E_{p1}$ (E_{p2}: 0)

$E_{c2} = E_{c1} + E_{p1} \rightarrow \frac{1}{2} m V_2^2 = \frac{1}{2} m V_1^2 + mgh$

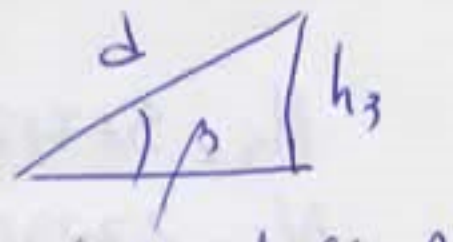
$$V_2 = \sqrt{V_1^2 + 2gh}$$

$$V_2 = \sqrt{617}\ m/s = 24,84\ m/s$$

b) $W_{FNC}^{2-3} = \Delta E_c + \Delta E_p + \Delta E_k$ (ΔE_k: 0)

$W_{fr}^{2-3} = E_{c3} - E_{c2} + E_{p3} - E_{p2}$ (E_{c3}: 0, E_{p2}: 0)

$W_{fr}^{2-3} = E_{p3} - E_{c2} = mgh_3 - \frac{1}{2} m V_2^2$



$h_3 = d \cdot \operatorname{sen}\beta$

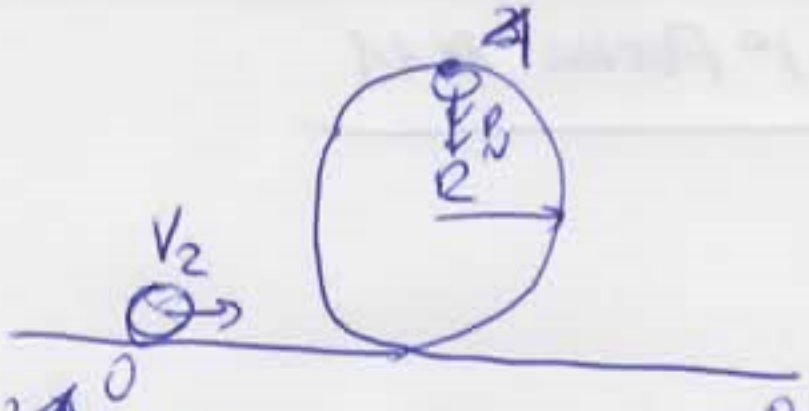
$$W_{fr}^{2-3} = mg\, d \operatorname{sen}\beta - \frac{1}{2} m V_2^2$$

$W_{fr}^{2-3} = -1117.061,97\ J \quad \beta = 10°$

$W_{fr}^{2-3} = -908.393,34\ J \quad \beta = 15°$

$W_{fr}^{2-3} = -704.550,65\ J \quad \beta = 20°$

d)



$W_{FNC} = \Delta E_C + \Delta E_P + \Delta E_E$ (0)

$0 = E_{C4} - E_{C2} + E_{P4} - E_{P2}$ (0)

$0 = \frac{1}{2} m V_4^2 - \frac{1}{2} m V_2^2 + mg\,2R$

$V_4^2 = V_2^2 + 2g\,2R \Rightarrow V_4^2 = V_2^2 + 4gR$

En el punto 4: $\Sigma F_r = m \cdot a_r$

$P + N_4 = m\frac{V_4^2}{R}$

$N_4 = m\frac{V_4^2}{R} - mg = m\left(\frac{V_4^2}{R} - g\right)$

$$\boxed{N_4 = m\left[\frac{(V_2^2 + 4gR)}{R} - g\right]}$$

$N_4 = 51712\ N$ 1

$N_4 = 27032\ N$ 2

$N_4 = 18805{,}33\ N$ 3

$N_4 = 14892\ N$ 4

$N_4 = 12224\ N$ 5

$N_4 = 10578{,}62 N$ 6

$N_4 = 9403{,}43\ N$ 7

$N_4 = 8522 N$ 8

$N_4 = 4836{,}44\ N$ 9

$N_4 = 4288\ N$ 10

$N_4 = 6839{,}27 N$ 11

$N_4 = 6465{,}33\ N$ 12

2)



a) Durante el choque se conserva $\vec{p}$

$$m \cdot V = mV' + M \cdot V_B$$

Nos Faltan 2 datos V', V_B → usamos la m de bloque

Luego del choque se conserva la E_m del bloque

$$E_{C1} = E_{K2}$$

$$\frac{1}{2} M V_B^2 = \frac{1}{2} K X^2 \rightarrow \boxed{V_B = \sqrt{\frac{K}{M}} \cdot X}$$

$\begin{cases} V_B = \\ V_B = \\ V_B = \\ V_B = \\ V_B = \\ V_B = \end{cases}$ $\begin{matrix} V_B = \\ V_B = \\ V_B = \\ V_B = 1 \\ V_B = \\ V_B = \end{matrix}$

c) $\boxed{V_B = \sqrt{\frac{K}{1\,kg}} \cdot 0{,}05\ m/s}$

$$V' = \frac{m \cdot V - M \cdot V_B}{m} = V - \frac{M}{m} V_B$$

$$V' = V - \frac{1\,kg}{0{,}05\,kg} \cdot 0{,}05 \frac{m}{s} \sqrt{K} \rightarrow \boxed{V' = V - \sqrt{K}\ m/s}$$ (6)

b) $\Delta E_C = \frac{1}{2} m V'^2 - \frac{1}{2} m V^2 \Rightarrow \boxed{\Delta E_C = \frac{1}{2} m \left(V'^2 - V^2\right)}$

d) $I = \Delta P = m \cdot M \cdot (V_B - 0) = M V_B \rightarrow \boxed{I = M V_B = M \sqrt{K} \times 0{,}05\ N \cdot s}$

$$\boxed{F_M = \frac{I}{\Delta t} = \frac{M \cdot 0{,}05 \sqrt{K}}{0{,}01\ s} = 5\sqrt{K}\ N}$$

e) $x(t) = A \operatorname{sen}(\omega t + \varphi_0)$

$A = 0{,}05\ m$

$\omega = \sqrt{\frac{k}{m}}$

$\varphi_0 = 0$ pues tomamos el movimiento en la posición de equilibrio

$$\boxed{x(t) = 0{,}05 \operatorname{sen}(\sqrt{k} \cdot t)\ m}$$

$$\boxed{v(t) = 0{,}05 \cdot \sqrt{k} \cdot \cos(\sqrt{k} \cdot t)\ m/s}$$

$$\boxed{a(t) = -0{,}05\ k \cdot \operatorname{sen}(\sqrt{k} \cdot t)\ m/s^2}$$

f) $\omega = 2\pi \cdot f \rightarrow f = \frac{\omega}{2\pi} = \frac{\sqrt{k/m}}{2\pi} \rightarrow \boxed{f = \frac{\sqrt{k}}{2\pi}\ Hz}$

$$\boxed{T = \frac{1}{f} = \frac{2\pi}{\sqrt{k}}\ s}$$

g) $E_c = \frac{1}{2} m v^2 = \frac{1}{2} M \cdot \left[\omega \cdot \sqrt{A^2 - x^2}\right]^2 = \frac{1}{2} M \cdot \omega^2 (A^2 - x^2)$

$E_k = \frac{1}{2} k x^2$